

Course: INFO 531: Data Warehousing and Analytics in the Cloud

Module/Week 9 - Week of October 17, 2022

Topic: Cloud Data Storage Models

Week 9 Assignment

Write a single-spaced, three-to-five-page summary of the learnings from the below article.

Understand data store models:

<https://learn.microsoft.com/en-us/azure/architecture/guide/technology-choices/data-store-overview>

The write-up should cover the 10 data storage offered by Azure:

1. Relational database management systems

Relational database stores two dimensional tables with rows and columns. An RDBMS typically implements a transactional consistent mechanism that conforms to the ACID model for modifying information.

Azure SQL database is always up-to-date fully managed database service which is managed by cloud.

Azure MySQL database is cost effective and easy to set up, operate and scale. Enjoy advanced security, same zone or zone redundant high availability, and a service level agreement of up to 99.9%.

Azure Database for PostgreSQL offers a service-level agreement (SLA) of up to 99.99 percent uptime, AI-powered performance optimization, and advanced security.

2. Key/value stores

A key/value data stores associate each data value with a unique key. Most key/value store only supports only simple insert, update and delete operations.

Key/value stores are highly optimized for applications performing simple lookups, but are less suitable if you need to query data across different key/value stores.

Key	Value
AAAAA	1101001111010100110101111...
AABAB	1001100001011001101011110...
DFA766	0000000000101010110101010...
FABCC4	1110110110101010100101101...

Opaque to data store

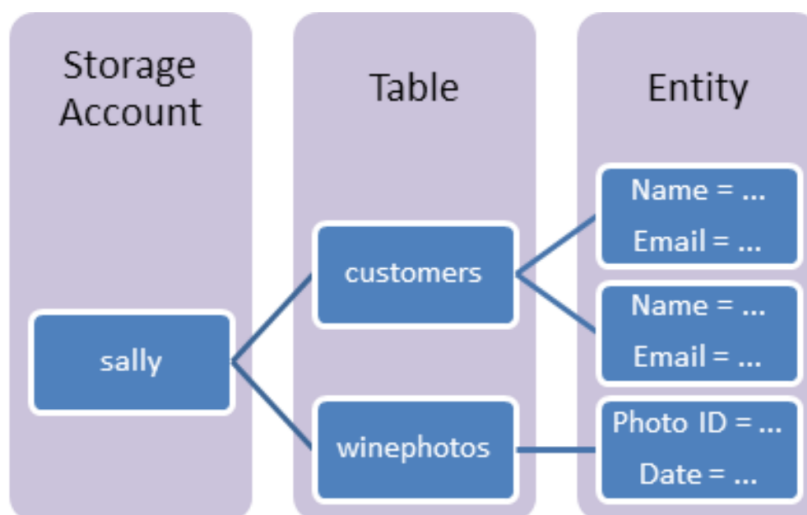
[Azure Cosmos DB](#) provides the API for Table for applications that are written for Azure Table storage and that need premium capabilities like:

- Guaranteed high availability.
- Automatic secondary indexing.

Azure table storage that stores non-relational data in the cloud, providing a key/ attribute store with a schemaless design.

ou can use Table storage to store flexible datasets like user data for web applications, address books, device information, or other types of metadata your service requires

Table storage contains the following components:



3. Document databases

A document database stores a collection of *documents*, where each document consists of named fields and data.

Typically, a document contains the data for single entity, such as a customer or an order.

Documents don't need to have the same structure.

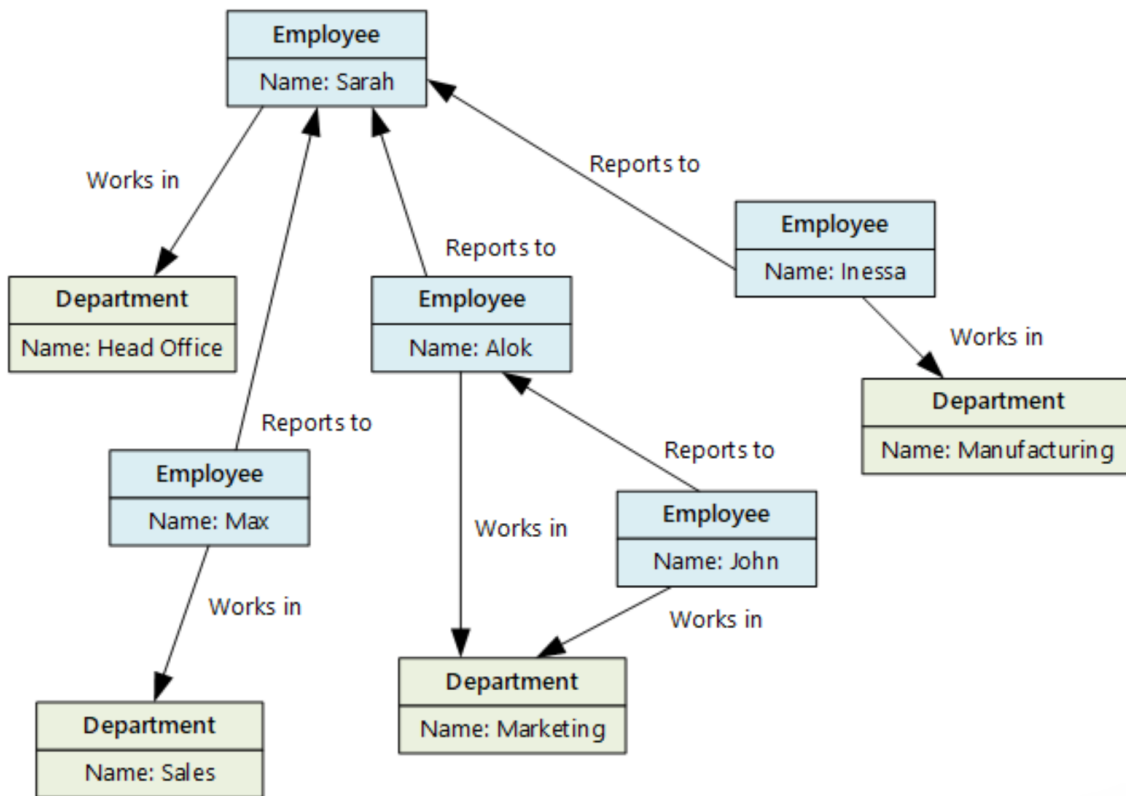
Key	Document
1001	<pre>{ "CustomerID": 99, "OrderItems": [{ "ProductID": 2010, "Quantity": 2, "Cost": 520 }, { "ProductID": 4365, "Quantity": 1, "Cost": 18 }], "OrderDate": "04/01/2017" }</pre>
1002	<pre>{ "CustomerID": 220, "OrderItems": [{ "ProductID": 1285, "Quantity": 1, "Cost": 120 }], "OrderDate": "05/08/2017" }</pre>

- Insert and update operations are common.

- No object-relational impedance mismatch. Documents can better match the object structures used in application code.
- Individual documents are retrieved and written as a single block.

4. Graph databases

A graph database stores two types of information, nodes and edges. Edges specify relationships between nodes.



Many graph databases provide a query language that you can use to traverse a network of relationships efficiently.

[Azure Cosmos DB](#) is the globally distributed, multi-model database service from Microsoft for mission-critical applications. It is a multi-model database and supports document, key-value, graph, and column-family data models. Azure Cosmos DB provides a graph database service via the Gremlin API on a fully managed database service designed for any scale.

5. Data analytics

Data analytics stores provide massively parallel solutions for ingesting, storing, and analyzing data.

- Historical data from multiple sources.
- Usually denormalized in a "star" or "snowflake" schema, consisting of fact and dimension tables.
- Usually loaded with new data on a scheduled basis.

6. Column-family databases

A column-family database organizes data into rows and columns. In its simplest form, a column-family database can appear very similar to a relational database, at least conceptually.

Figure 6-1: Column-family database structure. The figure shows two tables side-by-side. The left table is titled 'Column Family: Identity' and the right table is titled 'Column Family: Contact Info'. Both tables have a 'CustomerID' column and a 'Column Family' column. The 'Identity' table has three rows with customer IDs 001, 002, and 003, containing first and last names. The 'Contact Info' table has three rows with customer IDs 001, 002, and 003, containing phone numbers and email addresses.

CustomerID	Column Family: Identity
001	First name: Mu Bae Last name: Min
002	First name: Francisco Last name: Vila Nova Suffix: Jr.
003	First name: Lena Last name: Adamczyk Title: Dr.

CustomerID	Column Family: Contact Info
001	Phone number: 555-0100 Email: someone@example.com
002	Email: vilanova@contoso.com
003	Phone number: 555-0120

7. Search Engine Databases

A search engine database allows applications to search for information held in external data stores. A search engine database can index massive volumes of data and provide near real-time access to these indexes.

Indexes can be multi-dimensional and may support free-text searches across large volumes of text data.

Azure Cognitive Search is the only cloud search service with built-in AI capabilities that enrich all types of information to help you identify and explore relevant content at scale. Azure Cognitive Search also offers semantic search capability, which uses advanced machine learning techniques to understand user intent and contextually rank the most relevant search results.

Some search engines also support linguistic analysis that can return matches based on synonyms, genre expansions (for example, matching dogs to pets), and stemming (matching words with the same root).

8. Time series databases

Time series data is a set of values organized by time. Time series databases typically collect large amounts of data in real time from a large number of sources. Updates are rare, and deletes are often done as bulk operations.

Azure Time Series insight

Use time series data insights and interactive analytics to accelerate IoT analytics data use throughout your organization.

- Monitoring and event telemetry.
- Sensor or other IoT data.

9. Object storage

Object storage is optimized for storing and retrieving large binary objects (images, files, video and audio streams, large application data objects and documents, virtual machine disk images). Large data files are also popularly used in this model, for example, delimiter file (CSV),

10. Shared files

Sometimes, using simple flat files can be the most effective means of storing and retrieving information. Using file shares enables files to be accessed across a network.

Azure files

Azure Files shares can be mounted concurrently by cloud or on-premises deployments of Windows, Linux, and macOS. Azure Files shares can also be cached on Windows Servers with Azure File Sync for fast access near where the data is being used.

Differentiate between these data storages.

Relational databases :the data in an RDBMS must be normalized, which isn't appropriate for every data set.

Key/Value database:Data is accessed using a single key, like a dictionary.

No joins, lock, or unions are required.

No aggregation mechanisms are used.

Document: Documents don't need to have the same structure. Applications can store different data in documents as business requirements change.

Graph: Nodes are similar to table rows or JSON documents.

Relationships are just as important as nodes, and are exposed directly in the query language.

Shared files: Shared files are for sharing a file across the network.

Object:Object stores can manage extremely large amounts of unstructured data. Content is typically an asset such as a delimiter, image, or video file.Content must be durable and external to any application tier.

Time Series data: Time series data is a set of values organized by time. Time series databases typically collect large amounts of data in real time from a large number of sources.A timestamp is used as the primary key and sorting mechanism.

Search engine database: Full text search is required. Ad hoc self-service query is required. Data indexes from multiple sources and services. Queries are ad-hoc and can be complex.

Column family database: The real power of a column-family database lies in its denormalized approach to structuring sparse data. You can think of a column-family database as holding tabular data with rows and columns, but the columns are divided into groups known as *column families*. Each column family

holds a set of columns that are logically related together and are typically retrieved or manipulated as a unit. Unlike a key/value store or a document database, most column-family databases store data in key order, rather than by computing a hash.

Data analytics: Historical data is typically stored in data stores such as blob storage or [Azure Data Lake Storage Gen2](#), which are then accessed by Azure Synapse, Databricks, or HDInsight as external tables.

At the end of your essay, provide a summary table consisting of these storages. One column should show Azure services for each storage. Another column should show the data type for each storage. Another column should show the differences, similarities, or any other comments against each storage.

Data models	Definition	Azure Services	Data type	differences	similarities
Relational database management	Relational databases organize data as a series of two-dimensional tables with rows and columns. An RDBMS typically implements a transactionally consistent mechanism that conforms to the ACID (Atomic, Consistent, Isolated, Durable) model for updating information.	Azure SQL Database Azure database for MySQL Azure database for PostgreSQL Azure Database for MariaDB	Data is highly normalized. Database schemas are required and enforced. Many-to-many relationships between data entities in the database.	the data in an RDBMS must be normalized, which isn't appropriate for every data set.	
Key/Value	A key/value store associates each data value with a unique key. Most key/value stores only support simple query, insert, and delete operations.	Azure CosmosDB for Table Azure Table Storage	Each key is associated with a single value. There is no schema enforcement. No relationships between entities.	Data is accessed using a single key, like a dictionary. No joins, lock, or unions are required. No aggregation mechanisms	

				are used.	
Document	A document database stores a collection of <i>documents</i> , where each document consists of named fields and data. Documents are retrieved by unique keys.	Azure CosmosDB for NOSQL	Data can be managed in de-normalized way. Size of individual document data is relatively small. Each document type can use its own schema. Documents can include optional fields.	Documents don't need to have the same structure. Applications can store different data in documents as business requirements change. Data can be managed in de-normalized way.	Documents are retrieved by unique keys. Document data is semi-structured, meaning that data types of each field are not strictly defined.
Graph database	A graph database stores two types of information, nodes and edges. Edges specify relationships between nodes. Many graph databases provide a query language that you can use to traverse a network of relationships efficiently.	Azure CosmosDB for Apache Gremlin API	Nodes and relationships. Nodes are similar to table rows or JSON documents. Relationships are just as important as nodes, and are exposed directly in the query language. Composite objects, such as a person with multiple phone numbers, tend	Nodes are similar to table rows or JSON documents. Relationships are just as important as nodes, and are exposed directly in the query language.	

			to be broken into separate, smaller nodes, combined with traversable relationships		
Shared files	Sometimes, using simple flat files can be the most effective means of storing and retrieving information. Using file shares enables files to be accessed across a network.	Azure files	Files in a hierarchical set of folders. Accessible with standard I/O libraries.	Shared file shares a file across a network.	
Object	Object storage is optimized for storing and retrieving large binary objects (images, files, video and audio streams, large application data objects and documents, virtual machine disk images). Large data files are also popularly used in this model, for example, delimiter file (CSV),	Azure BLOB storage Azure Data Lake Storage	Data size is large Value is opaque	Object stores can manage extremely large amounts of unstructured data. Identified by key. Content is typically an asset such as a delimiter, image, or video file. Content must be durable and external to any application tier.	

Time Series data	Time series data is a set of values organized by time. Time series databases typically collect large amounts of data in real time from a large number of sources. Updates are rare, and deletes are often done as bulk operations.	Azure Time Series Insight	A timestamp is used as the primary key and sorting mechanism.	Time series databases typically collect large amounts of data in real time from a large number of sources. Updates are rare, and deletes are often done as bulk operations. Records are generally appended sequentially in time order. Deletes occur in bulk, and are made to contiguous blocks or records. Data is read sequentially in either ascending or descending time order, often in parallel.	
Search engine	A search engine database allows applications to search for information held in external data stores. A search engine	Azure Search	Semi-structured or unstructured text Text with reference to structured data	Full text search is required. Ad hoc self-service query is required.	Semi-structured or unstructured text Text with reference to structured data

	database can index massive volumes of data and provide near real-time access to these indexes.				
Column family database	A column-family database organizes data into rows and columns. In its simplest form, a column-family database can appear very similar to a relational database, at least conceptually.	Azure CosmosDB for Apache Casandra	Data is stored in tables consisting of a key column and one or more column families. Specific columns can vary by individual rows. Individual cells are accessed via get and put commands	The real power of a column-family database lies in its denormalized approach to structuring sparse data. You can think of a column-family database as holding tabular data with rows and columns, but the columns are divided into groups known as <i>column families</i>. Each column family holds a set of columns that are logically related together and are typically retrieved or manipulated as a unit	
Data analytics	Data analytics stores provide massively parallel solutions for ingesting,	Azure Data Lake Azure Data Explorer Azure data brick	Historical data from multiple sources. Usually denormalized	Historical data is typically stored in data stores such as blob storage or Azure Data	

	storing, and analyzing data		in a "star" or "snowflake" schema, consisting of fact and dimension tables. Usually loaded with new data on a scheduled basis. Dimension tables often include multiple historic versions of an entity, referred to as a <i>slowly changing dimension</i>.	Lake Storage Gen2 , which are then accessed by Azure Synapse, Databricks, or HDInsight as external tables.	
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The last paragraph of your write-up should list at least 5 key takeaways from the article.

- (1) Azure have provided different data models according to the type and size of data that are included in above chart.
- (2) The organization should use the data store according to the requirement.
- (3) They could also check the difference and similarities between each data stores and also they should compare the cost and scalability.
- (4) Azure have considered a small requirement of the organization including sharing a file, creating a graph to define the different data stores. It also satiate the small requirement of the organization.
- (5) The model is also defined by its structured and unstructured approach. Sometimes it supports both.

Upload your submission to the Week 9 Assignment Dropbox in D2L.